

Name _____

Section _____

MTH:2310 DISCRETE MATH
QUIZ 7
DUE: FRIDAY OCT. 31

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: (4 points)

Your ability to express your solutions is just as important as mathematical accuracy.

You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

Problem 1. (12 points) *Basic Modular Arithmetic*

(a) (6 points) Find $4^{-1} \pmod{23}$, if it exists. If it does not exist, exhaustively show why it can not exist.

(b) (6 points) Find $4^{-1} \pmod{26}$ if it exists. If it does not exist, exhaustively show why it can not exist.

Problem 2. (8 points) *Basic Modular Arithmetic*

Suppose a and b are integers such that $a \equiv 4 \pmod{23}$ and $b \equiv 12 \pmod{23}$.

- (a) (4 points) Find an integer $-46 \leq c \leq -24$ such that $c \equiv (b^2 - 4a) \pmod{23}$.

- (b) (4 points) Find an integer $23 \leq d \leq 45$ such that $a \cdot d \equiv b \pmod{23}$

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Problem 3. (12 points) *Basic Modular Arithmetic*

Evaluate the following expressions.

(a) (6 points) $\left(3^{99} + (-135) \cdot (35)\right) \bmod 13$

(b) (6 points) $\left((5^3 \bmod 23)^2 \bmod 31\right) \bmod 4$

Problem 4. (12 points) *Primes and Greatest Common Divisors*

(a) (6 points) Use Trial Division to determine whether 221 is prime or composite. Show every step.

(b) (6 points) Use Trial Division to determine whether 211 is prime or composite. Show every step.

Problem 5. (8 points) *Primes and Greatest Common Divisors*

Find the prime factorization of the following integers.

(a) 504

(b) $\frac{16!}{5!11!}$